

Bo-Jian (Eddie) Ho

Data Engineer

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With 2 years of experience in data engineering, specializing in building scalable data pipelines and ETL workflows using Python and GCP. Proficient in designing data lakehouse architectures, distributed systems, and implementing machine learning-driven solutions. Committed to advancing data-driven products, with an interest in applying skills to geospatial data and GIS operations.

Employment history

Data Engineer, Aug 2025 – Present

Million Software Inc., San Francisco, CA

- Engineered CDC pipelines using *Debezium*, *Apache Kafka*, and *ClickPipe* to sync **200TB+ historical data** and **1TB+ daily data** with zero downtime; reducing end-to-end data processing time by **70%**, and enhancing security engineering controls for data integrity
- Architected scalable data lakehouse infrastructure using *Apache Iceberg* on AWS S3, *PlanetScale* (OLTP), and *ClickHouse* (OLAP), enabling high-throughput analytics and time-travel queries across **200TB+** of data, incorporating privacy infrastructure and security software
- Developed production-grade REST APIs serving internal and customer-facing analytics at **200K+ daily requests**, achieving **sub-second response times** via *ClickHouse* materialized views, caching, and query optimization, with attention to data security best practices
- Reduced cloud costs by **\$2K/month** by migrating to managed services, utilizing ephemeral processing resources, optimizing S3 storage tiering, right-sizing auto-scaling, and refactoring pipeline queries to cut redundant compute
- Authored lightweight data validation suites that caught schema drift early, reducing incident recovery effort and sustaining report accuracy

Member of Technical Staff, Dec 2024 – Aug 2025

Chima Inc., San Francisco, CA

- Built automated geocoding pipeline using Google Maps Geocoding API to standardize **2M+ historical** and **10K+ monthly** address records, **achieving 95% match rate** and enabling downstream geospatial analysis
- Engineered *ETL* pipelines to ingest and process **10M+ geospatial records** from OpenStreetMap and municipal GIS databases, transforming vector data into optimized formats for *BigQuery Geo* queries
- Developed end-to-end *RAG* system to enable semantic search for AI agents, architecting the full *Data Pipeline* from query embedding generation using sentence transformers to vector similarity search with *Turbopuffer* and model-based reranking
- Optimized with embedding caching, batch processing, and hybrid retrieval combining semantic and keyword search using *LangChain* and *LangGraph* to achieve consistent **sub-500ms latency** for concurrent user queries at scale
- Implemented incremental indexing pipeline and S3 *One-Zone* cache storage for selective re-embedding of modified code sections, reducing re-indexing time by 50% and achieving **millisecond retrieval latency**
- Established evaluation using LLM-as-a-judge methodology to validate semantic search performance against traditional grep-based search, demonstrating a **70% win rate in relevance and accuracy** across diverse query patterns

Data Engineer, May 2024 – Nov 2024

Lesso America Inc., Los Angeles, CA

- Built ETL pipeline monitoring and alerting system using GCP Monitoring, custom data quality tests, and Airflow sensors to detect volume anomalies, schema drift, and errors; cutting incident resolution time to under 4 hours
- Implemented automated invoice scanning feature using Mistral OCR, integrated into the internal finance system, eliminating manual data entry and enabling accounting clerks to process 30% more invoices per day, leveraging Python and GCP services
- Supported integration of Geographical Data and GIS concepts in data governance and reporting pipelines
- Established data governance policies including role-based access controls and column-level masking in BigQuery and Data Catalog to ensure compliance with data privacy requirements, secure handling of sensitive business information, and proper database administration

Student Engineer, Sep 2023 – Mar 2024

UC San Diego, San Diego, CA

- Developed distributed data transformation pipelines using *Apache Spark* with *Java*, processing large-scale datasets through *MapReduce*
- Established structured logging framework using *Python* logging module and JSON formatters with contextual metadata tagging, improving debugging workflows and operational observability across services
- Optimized backend database queries through strategic indexing on frequently-accessed columns and query rewriting, achieving **~30% improvement** in average query execution time across production workloads
- Wrote unit and integration tests using *pytest* across core services, improving code coverage and reducing production errors

Education

Bachelor of Science in Data Science, Sep 2021 – Mar 2024

University of California-San Diego, San Diego, CA

- GPA 3.8; Minor in Cognitive Science; Member of Data Science Student Society and Computer Science & Engineering Society

Skills

Python, SQL, Typescript, Java, React, Machine Learning, Distributed System, ETL, PySpark, Apache Airflow, Apache Kafka, Apache Iceberg, DBT, AWS, GCP, MySQL, MongoDB, BigQuery, Data Modeling, Geographical Data, GIS, Vector Data

Links

GitHub: github.com, Personal Website: eddieho.xyz, Ami: ami.dev, Same: same.new, Langflow: langflow.org